

An Empirical Analysis of Distributional Effects in Learning-Based Routing Systems

Daniela Thyssens^{1*}, Tim Dornedde¹, and Lars Schmidt-Thieme¹

Information Systems and ML Lab, University of Hildesheim, Hildesheim, Germany
{thyssens, dornedde, schmidt-thieme}@ismll.uni-hildesheim.de

Abstract. Learning-based approaches for routing and transportation problems are increasingly considered as components of autonomous decision-making systems. However, their empirical performance is typically evaluated on independently sampled synthetic instances, a setting that differs substantially from real-world deployment scenarios where environments exhibit recurring structure. In this paper, we present a systematic empirical study of how instance distribution design and evaluation protocols influence the observed behavior of neural routing solvers. Focusing on Traveling Salesman and Capacitated Vehicle Routing Problems, we compare learning-based methods with classical optimization solvers under controlled distributions that vary the degree of structural recurrence. Our results show that learning-based routing systems benefit significantly from structured and recurring environments, substantially narrowing, and in some cases reversing, the performance gap to classical solvers under matched runtime budgets. At the same time, we identify limitations in out-of-distribution generalization and discuss conditions under which these gains diminish. These findings highlight the importance of evaluation protocols that reflect the operational characteristics of autonomous transportation systems and provide insights into when and why learning-based decision models are effective. The study contributes diagnostic evidence for understanding the role of environmental structure in data-driven decision-making.

Keywords: Learning-Based Routing · Autonomous Transportation · Distributional Effects · Evaluation Methodology

1 Introduction

Learning-based approaches are increasingly used to support decision-making in combinatorial optimization problems arising in transportation and logistics. Neural Combinatorial Optimization (NCO), particularly in the context of routing problems, aims to learn a parametrized policy that maps problem instances to near-optimal solutions, reducing manual algorithm design effort and enabling fast inference compared to traditional optimization methods. To train such models, problem instances are typically sampled from simple synthetic generators, for example using uniformly distributed customer coordinates and demands in the Vehicle Routing Problem (VRP) [2]. While this convention has enabled rapid empirical

* Corresponding author: thyssens@ismll.uni-hildesheim.de

progress, it removes a key ingredient of data-driven learning: recurring structure across related instances. In many autonomous transportation and logistics settings, learning-based decision systems are often deployed repeatedly within environments that exhibit substantial structural regularity. Customer locations, service regions, and operational constraints often persist over time, allowing models to exploit recurring spatial and relational patterns. In contrast, common evaluation protocols for neural routing models rely on independently generated synthetic instances, effectively eliminating such environmental structure. This discrepancy raises the question of whether current benchmarks accurately reflect the capabilities and limitations of learning-based decision systems in structured deployment settings. To investigate this issue, we study a subsampling-based evaluation setting in which both training and test instances are drawn from a shared **Base Node Distribution**. Instead of generating each instance independently, problem instances are constructed by subsampling from a fixed underlying set of nodes (e.g., customer locations). Figure 1 illustrates the contrast to conventional independent sampling. Fixing the underlying instance distribution makes explicit how learning-based routing systems acquire and exploit structure under repeated deployment conditions, reframing neural routing as a distribution-aware learning problem. This controlled structural recurrence enables a focused analysis of when neural solvers benefit from distributional alignment, without relying on real-world datasets. We examine this setting on two canonical routing problems—the Traveling Salesman Problem (TSP) and the Capacitated Vehicle Routing Problem (CVRP). Three representative neural routing methods (POMO [15], BQ [6], NeuOpt [17]) are retrained under distribution-aligned and conventional regimes and compared to strong classical meta-heuristic solvers (LKH-3 [9], HGS-CVRP [23]). In summary, we make the following contributions:

- We conduct a systematic empirical analysis of learning-based and classical routing solvers under controlled instance distributions with varying degrees of structural recurrence.
- We show that shared distributional structure can substantially influence the in-distribution performance of neural routing methods, narrowing the gap to strong classical solvers under matched runtime budgets.
- We analyze limitations under distributional shifts and discuss implications for the evaluation of learning-based decision systems in structured transportation settings.

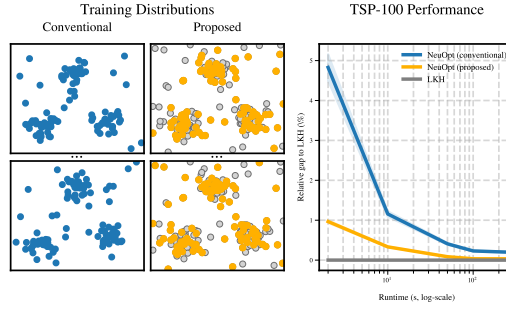


Fig. 1: **Learning Under Shared Base Distributions.** Left: conventional training instances (blue) & training instances (yellow) sampled from a shared base node distribution (grey). Right: Performance on test instances drawn from the base distribution.

2 Related Work

NCO Data Distributions. NCO assumes the existence of an underlying distribution over the instances within a given problem class [14, 3], since most combinatorial problems are NP-hard, such that no single solver can be efficient across all possible instances. In routing tasks, instance attributes such as node coordinates and demands are typically sampled independently and uniformly at random [19], resulting in minimal structural complexity. For evaluating the capabilities of a learned system however, the instance distribution must exhibit sufficient structure to be discoverable [26]. While recent work has explored more diverse or structured distributions [24, 26, 4], most prior work still generates problem instance attributes independently across instances. Our work complements this line of research by inducing recurrence through subsampling from a shared base node distribution, providing a complementary and solver-agnostic evaluation setting.

Neural Methods for Routing Problems. NCO approaches for routing tasks fall broadly into two classes: *constructive* methods, which build solutions sequentially, and *improvement*-based methods, which refine existing ones. Constructive methods typically use attention-based encoders and autoregressive decoders to select the next node [14, 15, 12, 6, 16], sometimes combined with search procedures such as beam search or sampling [5, 6]. Hybridizations with heuristic concepts, e.g., dynamic programming, large neighborhood search, or decomposition, have further improved accuracy [11, 7, 25]. Improvement-based solvers, by contrast, learn to modify or select local search moves [8, 17, 24, 17, 18]. Despite architectural differences, both paradigms remain sensitive to the structure of the instance distribution on which they are trained. We therefore evaluate representative methods from both categories under the proposed subsampling-based evaluation setting to reveal how distributional structure influences their performance and robustness.

Neural Methods and Generalization. Recent studies have examined the generalization behavior of learned routing solvers, exploring scale conditioning, meta-learning, or test-time adaptation [13, 26, 10]. Most prior works assess out-of-distribution (OOD) performance by varying instance size or sampling distribution. In contrast, we focus on **in-distribution learning**, i.e., how effectively a model fits the distribution (Base Node Distribution) it was trained on. Given the NP-hard nature of these problems, no solver performs uniformly well across all instance types, making in-distribution performance a more principled indicator of the model’s learning behavior. At the same time, robustness to OOD scenarios is undoubtedly important for practical applications, thus we demonstrate in controlled experiments that subsampling does not eliminate OOD generalization, but instead clarifies where robustness persists and where it breaks down.

3 Method

3.1 Problem Formulation

We consider NP-hard binary combinatorial optimization problems, and more specifically routing problems, over graphs. Let $G = (V, E)$ be a graph, where V are the vertices or nodes and $E \subseteq V \times V$ are the edges of G . The optimization problem consists of a set $\Omega_G \subseteq \{0, 1\}^N$ of *feasible solutions* and an *objective function* $f_G: \Omega_G \rightarrow \mathbb{R}$ and the task is to find an optimal feasible solution $x = (x_1, \dots, x_N) \in \Omega$, by minimizing the objective function:

$$\min_{x \in \Omega_G} f_G(x). \quad (1)$$

For $x \in \{0, 1\}^N$, the elements x_i are also referred to as the binary decision variables. While for the TSP, the task is to select a subset of edges such that the Hamiltonian cycle with minimum distance is found [1], the task in the VRP is to find a feasible solution that consists of a set of feasible routes $r_k = (x_{k,1}, \dots, x_{k,R_k})$, which is a sequence of nodes always starting and ending at the depot node, such that $x_{k,1} = 0$ and $x_{k,R_k} = 0$ and R_k denotes the length of the route [21].

3.2 Subsampling Framework

Machine Learning (ML) methods are particularly adept at uncovering patterns and structures inherent in datasets drawn from a specific underlying distribution. As a result, instance distribution design becomes a central factor when analyzing the behavior of learning-based decision systems under repeated and structured operating conditions. To study this effect in routing problems, we consider a subsampling-based evaluation setting in which training and test instances are drawn from a shared *Base Node Distribution*. Formally, we define a base node distribution $\mathcal{G}_{\text{base}}$ that is sampled from an underlying distribution $\mathcal{D}$:

$$\mathcal{G}_{\text{base}} \sim \mathcal{D} \quad (2)$$

where $\mathcal{D}$ represents the coordinate and demand distributions associated with the N_{base} customer nodes $\{V_b\}_{b=0}^{N_{\text{base}}}$ sampled for the base node distribution. In the case of the CVRP, node V_0 in $\mathcal{G}_{\text{base}}$ is the designated depot node, which remains fixed for subsampled CVRP problem instances. Given the base node distribution $\mathcal{G}_{\text{base}}$ we sample nodes through f_{sample} to form an instance G_i of problem-size N :

$$G_i = V_1, V_2, \dots, V_N \sim f_{\text{sample}}(\mathcal{G}_{\text{base}}) \quad \text{for } i = 1, \dots, L \quad (3)$$

where L denotes the number of instances and f_{sample} corresponds to the uniform sampling function, such that $G_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{G}_{\text{base}}$. For CVRP, the depot node V_0 is always included in each subsampled instance G_i . Finally, we define the train and test sets associated with the Base Node Distribution $\mathcal{G}_{\text{base}}$ as

$$\mathbf{G}_{\text{train}} := \{G_i\}_{i=1}^{L_{\text{train}}}, \quad \mathbf{G}_{\text{test}} := \{G_i\}_{i=L_{\text{train}}+1}^{L_{\text{train}}+L_{\text{test}}}.$$

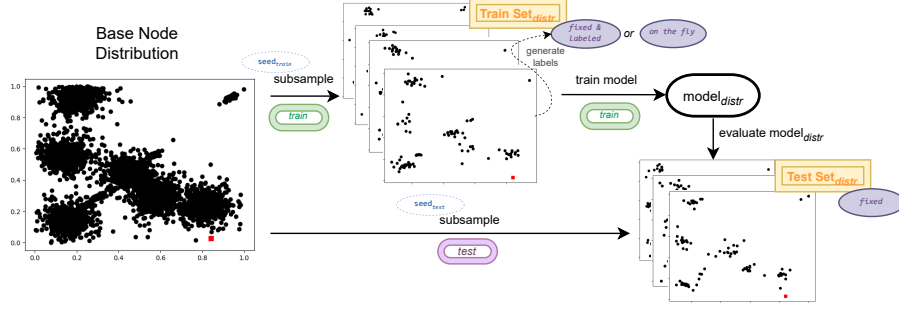


Fig. 2: **Subsampling-Based Evaluation Setting.** A Base Node Distribution $\mathcal{G}_{\text{base}}^X$ is fixed (left). Training and test instances are generated by subsampling from this shared set (with a fixed depot), inducing controlled structural recurrence across instances.

In our experiments, we instantiate $\mathcal{D}$ using four standard routing distributions (Uniform, Rotation, Explosion, and Uchoa (X)). For each choice of $\mathcal{D}$, we sample a single Base Node Distribution $\mathcal{G}_{\text{base}}$, which remains fixed across training runs and defines the corresponding train/test splits used to evaluate performance under different structural regimes. To support both reinforcement learning and supervised training pipelines, we employ a unified dataset interface that generates either on-the-fly subsampled batches (for RL) or fixed labeled datasets (for imitation learning). For supervised methods, training labels are obtained using LKH-3 [9] for TSP and HGS-CVRP [23] for CVRP. Figure 2 illustrates the subsampling scheme for an exemplary Base Node Distribution with $N_{\text{base}} = 10,000$ nodes. The base distribution $\mathcal{G}_{\text{base}}^X$ follows the structured coordinate patterns introduced in [22], including clustered customer locations. Training and test instances are generated by subsampling from this fixed base set using different random seeds. Reinforcement-learning models draw subsamples on the fly during training, whereas supervised models rely on pre-constructed labeled datasets. After training on subsampled instances, models are evaluated on a fixed subsampled test set $\mathbf{G}_{\text{test}}^X$. By fixing the underlying instance space while varying sampled problem instances, this setting enables a focused analysis of how learning-based routing systems internalize and exploit recurring structural patterns.

4 Experiments

We evaluate the subsampling setting across four underlying instance distributions for both TSP and CVRP. Three representative NCO models, POMO [15], BQ [6], and NeuOpt [17], are trained under both the conventional independent-instance regime and the proposed subsampling framework. SGBS-EAS [5], which operates on top of a trained POMO policy, is included only at evaluation time.

4.1 Experimental Setup

We follow the standard generative definitions of the *Rotation* [4], *Explosion* [4], *X* [22], and *Uniform* [14] distributions. For each underlying distribution, we construct a fixed Base Node Distribution $\mathcal{G}_{\text{base}}$ of size $N_{\text{base}} \in \{200, 10\,000\}$ from which problem instances of size $N = 100$ are obtained via uniform subsampling. We follow standard practice in adopting $N = 100$ for all problem instances, and consider two base sizes (200 and 10 000) to reflect low- and high-diversity subsampling regimes. Table 1 summarizes the generated datasets. For completeness, we also include test sets sampled directly from the underlying Explosion and X distributions. These sets are used only for OOD comparisons (Section 4.4).

Table 1: Subsampled test sets used in our evaluation. N is the problem size, N_{base} is the size of the Base Node Distribution from which instances are subsampled. Each test set contains 128 instances.

Dataset	N	N_{base}	Problem Class	Distribution #	Test Instances
$\mathbf{G}_{200}^{\text{Unif}}$	100	200	TSP, CVRP	Uniform [14]	128
$\mathbf{G}_{10k}^{\text{Unif}}$	100	10000	TSP, CVRP	Uniform [14]	128
$\mathbf{G}_{200}^{\text{X}}$	100	200	TSP, CVRP	X_C, X_{RRC0} [22]	128
$\mathbf{G}_{10k}^{\text{X}}$	100	10000	TSP, CVRP	X_C, X_{RRC0} [22]	128
$\mathbf{G}_{200}^{\text{Exp}}$	100	200	TSP, CVRP	Explosion [4]	128
$\mathbf{G}_{10k}^{\text{Exp}}$	100	10000	TSP, CVRP	Explosion [4]	128
$\mathbf{G}_{200}^{\text{Ro}}$	100	200	TSP, CVRP	Rotation [4]	128
$\mathbf{G}_{10k}^{\text{Ro}}$	100	10000	TSP, CVRP	Rotation [4]	128

Training Protocol. For each underlying distribution and problem setting, all NCO models are trained both under the proposed subsampling scheme and under the conventional independent-instance regime, using the same number of training instances for a fair comparison.

Evaluation and Runtime Budgets. Performance is reported as average optimality gap (%) with respect to the reference OR solver under matched wall-clock budgets T_{max} . Constructive models use their standard decoding pipelines, including beam search (BQ) followed by local improvement via simulated annealing [20]. Improvement-based models allocate the evaluation budget to their internal search operations (e.g., rollouts or Monte-Carlo steps), so T_{max} reflects practical runtime conditions rather than equalizing internal step counts. The experimental code is publicly available¹.

¹ <https://github.com/danielathyssens/distributional-dependent-routing>

Table 2: Conventional vs. subsampled training on CVRP test sets $\mathbf{G}_{200}^j$ ($j \in \{\text{Ex}, \text{Ro}, \text{X}, \text{Unif}\}$). Entries report average optimality gap (%) to HGS at $T_{\max} = 10\text{s}$. Percent improvements of the subsampled variant are shown in parentheses.

	Conventional			Subsampled			Lift
	BQ	POMO	SGBS	BQ _{sub}	POMO _{sub}	SGBS _{sub}	
$\mathbf{G}_{200}^{\text{Ex}}$	6.732	3.321	5.193	3.777 (43.9%)	2.585 (22.2%)	3.739 (28.0%)	31.3%
$\mathbf{G}_{200}^{\text{Ro}}$	1.717	2.122	3.699	1.670 (2.7%)	1.953 (8.0%)	2.634 (28.8%)	13.2%
$\mathbf{G}_{200}^{\text{X}}$	0.841	4.486	5.94	0.447 (46.9%)	1.554 (65.4%)	2.124 (64.2%)	58.8%
$\mathbf{G}_{200}^{\text{Unif}}$	0.880	1.949	2.509	0.922 (-4.8%)	1.756 (9.9%)	2.398 (4.4%)	3.2%

4.2 In-Distribution Performance Under Distribution-Aligned Training

We first assess the impact of the proposed framework when training and testing are aligned through a shared Base Node Distribution. Specifically, we evaluate conventionally trained & subsampling-trained models on subsampled test sets. Table 2 reports results on $\mathbf{G}_{200}^j$ for CVRP. Across all structured distributions (Explosion, Rotation, X), subsampling yields substantial improvements for all three neural baselines, with constructive methods (BQ, POMO) benefiting the most: relative gains range from 22–65%, and reach **+58.8%** on the clustered X distribution. Improvements are modest on Uniform data, consistent with the absence of recurring geometric patterns. Overall, these results support that subsampling introduces a beneficial inductive bias by exposing models to repeated spatial and relational patterns during training. Figure 3 shows analogous trends for TSP. The effect is less pronounced than for CVRP but remains consistent: BQ_{sub} and POMO_{sub} achieve lower gaps on the structured distributions and maintain an advantage as runtime increases. Uniform instances again show minimal differences. Runtime budgets in Fig. 3 are intentionally kept small to emphasize the constructive nature of the evaluated baselines. Rotation exhibits highly variable performance for some constructive baselines (notably POMO), so we omit it from the bar plot and discuss it qualitatively; POMO can exceed 50% gap, while BQ trained with subsampling achieves very small gaps ($\approx 0.1\%$).

Search-Based Methods. As search-based NCO methods typically benefit from longer runtimes, Figure 4 evaluates them at 240 s. Subsampling consistently improves SGBS, while NeuOpt_{sub} shows a short warm-up before converging to stronger solutions. This suggests that search-based models also exploit recurring structure, though improvements emerge later in the search process than in constructive models.

Runtime Trajectories. To complement the short-horizon analysis, Figure 5 reports full 240s runtime trajectories for both constructive and search-based models.

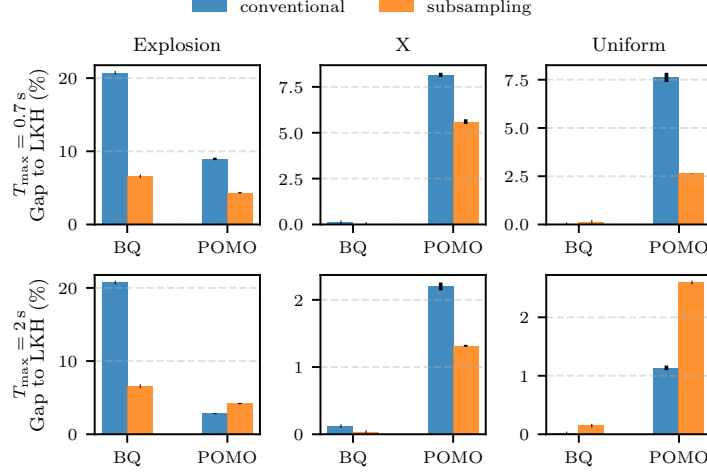
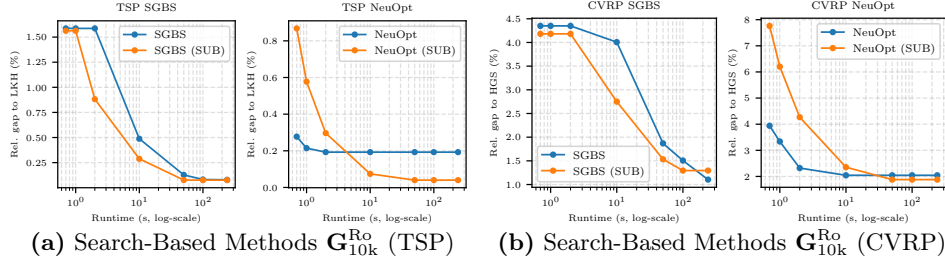


Fig. 3: Effect of subsampling on TSP test sets ($N_{\text{base}}=200$). Bars show average relative gap (%) to LKH at different time budgets. Smaller is better.



(a) Search-Based Methods $\mathbf{G}_{10k}^{\text{Ro}}$ (TSP)

(b) Search-Based Methods $\mathbf{G}_{10k}^{\text{Ro}}$ (CVRP)

Fig. 4: Effect of subsampling on NCO search methods for Rotation. Curves report average relative gap (%) to HGS or LKH as a function of runtime for $T_{\max} = 240s$.

Across CVRP and most TSP settings, subsampled variants maintain lower gaps over the majority of the runtime budget. The trajectories of POMO and SGBS closely align, reflecting POMO initialization for SGBS, yet subsampled training yields clear improvements for both in CVRP and in the early stages of TSP. On structurally complex distributions such as Explosion, early-runtime gains ($\leq 2s$) are particularly pronounced for constructive models, even when later-long-horizon performance is similar or slightly favors the conventional variant. Overall, these patterns rein-

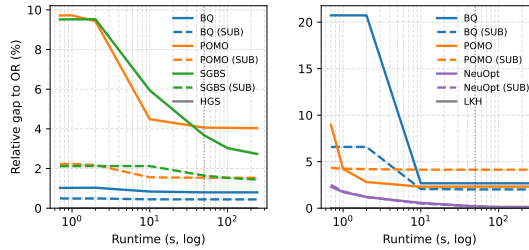


Fig. 5: Runtime trajectories at $N_{\text{base}}=200$ for CVRP $\mathbf{G}_{200}^{\text{X}}$ (left) and TSP $\mathbf{G}_{200}^{\text{Ex}}$ (right). Shaded areas denote $\pm 1\sigma$.

force that subsampling strengthens the constructive stage of NCO models and accelerates early convergence, which propagates into improved search trajectories for most distributions and provides a model-agnostic data design principle for enhancing in-distribution performance. Together, these results illustrate how controlled exposure to recurring structure shapes the learning dynamics of data-driven routing systems.

4.3 Comparison to Classical OR solvers

Table 3 compares subsampling-trained models to classical OR solvers under equal wall-clock budgets, focusing on structured in-distribution settings. For CVRP, subsampling substantially increases competitiveness relative to HGS. On the X distribution, both POMO_{sub} and SGBS_{sub} achieve negative gaps (-3.94% and -4.23%), exceeding HGS within the same runtime. On other base node distributions, subsampling narrows the gap to HGS. We also observe that models trained on the larger base node distribution ($N_{\text{base}} = 10\,000$) generally outperform their $N_{\text{base}} = 200$ counterparts when trained through RL, indicating that richer base distributions provide beneficial structural diversity for RL-based methods. For TSP, neural models trained under the subsampling scheme almost close

Table 3: Comparison of NCO models trained on base node distributions with classical OR metaheuristics. Negative gaps (bolded) indicate cases where learned models outperform LKH or HGS under the same runtime budget ($T_{\text{MAX}} = 50s$).

(a) TSP							(b) CVRP							
Gap (%)		ML				OR	Gap (%)		ML				OR	
Dataset		POMO _{sub}	BQ _{sub}	SGBS _{sub}	NeuOpt _{sub}	LKH	Dataset		POMO _{sub}	BQ _{sub}	SGBS _{sub}	NeuOpt _{sub}	LKH	HGS
N _{base} = 200	Explosion	4.141	2.018	0.863	0.190	0.000	N _{base} = 200	Explosion	2.554	3.319	2.800	3.190	1.514	0.000
	Rotation	3.462	0.056	17.356	1.248	0.000		Rotation	1.945	1.658	1.648	12.079	1.291	0.000
	Uniform	2.594	0.131	1.683	0.085	0.000		Uniform	1.746	0.897	2.290	8.255	1.599	0.000
	X	1.203	0.028	3.538	8.967	0.000		X	1.536	0.444	1.644	10.020	0.165	0.000
N _{base} = 10k	Explosion	1.023	0.139	0.665	0.536	0.000	N _{base} = 10k	Explosion	1.480	2.998	1.330	7.033	1.639	0.000
	Rotation	0.659	0.103	0.076	0.040	0.000		Rotation	2.568	1.441	1.533	1.879	1.126	0.000
	Uniform	1.219	0.110	0.045	0.048	0.000		Uniform	3.440	1.681	2.990	5.402	1.691	0.000
	X	2.393	0.099	0.794	2.035	0.000		X	-3.936	3.052	-4.225	-1.966	0.218	0.000

the gap to LKH; $\text{NeuOpt}_{\text{sub}}$ yields the strongest overall performance (gaps of 0.04% – 0.08%), while BQ_{sub} performs best on the X distribution with a gap of 0.028% .

Interpretation. Overall, aligning training and test distributions allows neural solvers to close much of the gap to classical metaheuristics, and on structured CVRP even surpass them. OR solvers remain stronger in broad, distribution-agnostic settings, but the benefits of subsampling are clearest in structured,

Table 4: In-distribution (ID) vs. OOD performance at $T_{\max} = 50$ s for CVRP and TSP. We compare conventional training with subsampling-trained models with $N_{\text{base}} = 200$. ID uses Base Node-subsampled test sets, OOD uses full X- or Explosion- sampled instances. Gaps are relative to the OR solver; $\Delta = \text{Gap}_{\text{out}} - \text{Gap}_{\text{in}}$; *Wins* is the percentage of OOD instances beating the OR baseline.

Problem Model		Uchoa				Explosion			
		Gap _{in}	Gap _{out}	Δ Wins (%)		Gap _{in}	Gap _{out}	Δ Wins (%)	
CVRP	BQ	0.797	-2.829	-3.627	59	6.620	6.813	0.193	0
	SGBS	3.673	-1.479	-5.152	50	3.008	5.117	2.109	0
	BQ _{sub}	0.444	-0.567	-1.011	45	3.319	6.110	2.790	0
	SGBS _{sub}	1.644	8.650	7.006	13	2.800	9.693	6.893	0
	HGS	0.000	0.000	0.000		0.000	0.000	0.000	
TSP	BQ	0.082	0.047	-0.035	19	2.674	2.973	0.299	0
	SGBS	1.156	0.958	-0.198	0	0.970	0.867	-0.103	7
	BQ _{sub}	0.028	1.513	1.485	0	2.018	2.521	0.503	5
	SGBS _{sub}	3.538	19.734	16.196	0	11.893	14.368	2.474	0
	LKH	0.000	0.000	0.000		0.000	0.000	0.000	

domain-aligned regimes. This highlights the value of evaluating NCO methods on distributions with more intricate geometric structure.

4.4 Out-of-Distribution Generalization

While the primary focus of this work is improving in-distribution performance (Section 4.2), we additionally assess generalization to full Uchoa- and Explosion-sampled instances from the same distribution family $\mathcal{D}$. Table 4 reports in-distribution (ID) results on subsampled test sets ($\mathbf{G}_{200}^{\text{Ex}}$, $\mathbf{G}_{200}^{\text{X}}$) and corresponding OOD performance. OOD behaviour is strongly architecture- and training-dependent, revealing differing trade-offs between exploiting structural recurrence and maintaining robustness under shift. Constructive models remain stable: BQ_{sub} exhibits very small OOD shifts ($\Delta \approx 0.1$ – 0.6) and even achieves negative gaps on full Uchoa CVRP, indicating that the structural bias induced by subsampling transfers beyond the exact base-node setting. On Explosion, BQ_{sub} does not improve over its own ID score but still outperforms the conventionally trained variant. Search-based RL models show higher OOD sensitivity. SGBS experiences notable degradation (e.g., TSP Uchoa: $\Delta = +16.2\%$) despite sizeable ID gains, reflecting a trade-off between leveraging recurrence during training and maintaining robustness under shift. Similar patterns hold for POMO and NeuOpt; full results appear in the anonymous repository. In summary, subsampling improves ID performance without inducing brittle overfitting for constructive models,

while highlighting architecture-dependent robustness limits for RL-driven search methods.

5 Discussion and Conclusion

This work examined how instance distributions shape the performance of neural and classical routing solvers—an aspect that remains underexplored in standard NCO evaluation practice. We studied a subsampling-based evaluation setting in which training and testing are aligned through a shared Base Node Distribution, enabling models to repeatedly encounter structured geometric patterns that conventional independent sampling tends to dilute. Across TSP and CVRP, distribution-aligned training consistently improves in-distribution performance. For CVRP, this alignment substantially narrows the gap to well-established classical meta-heuristics and, under structured distributions and limited runtime budgets, can even surpass them. For TSP, subsampling primarily accelerates convergence and reduces optimality gaps, while classical solvers remain competitive in broader, distribution-agnostic regimes. Out-of-distribution experiments further show that subsampling does not inherently induce brittle overfitting. Constructive supervised models such as BQ maintain stable generalization behavior, whereas RL-trained search methods exhibit greater sensitivity to distributional shift, highlighting architecture-dependent robustness trade-offs. Overall, our findings suggest that learning-based routing systems are inherently distribution-sensitive: their effectiveness depends strongly on the alignment between training distributions and the structural characteristics of deployment instances. These results underscore the importance of principled instance distribution design when evaluating neural combinatorial optimization methods and motivate further study of distribution-aware training strategies under structured operational conditions.

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